

VI Semester B.E. Examination
(Common to E&C/E&C-II(Industry Integrated))
Automotive Electronics (22EECC305/22EEIC316)

Duration: 3 hours**Max. Marks: 100**

Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II and any ONE full question from UNIT-III.

UNIT-I

- 1
 - a. The engine ECU is detected the malfunctioning of MAF sensor. How engine control unit can work effectively with other existing sensor information for calculating the mass air flow rate in engine management system. (08marks)
 - b. Demonstrate V-design model applied to embedded systems software development process. (06marks)
 - c. Suggest and explain the suitable sensor and actuator, which are used for closed loop operation of an engine control system to maintain the desired air/fuel ratio. (06marks)
- 2
 - a. The vehicle running at a very high speed as per the driver's input, and suddenly the engine encounters a rapid rise in cylinder pressure. Suggest and explain a suitable control algorithm to overcome this condition. (08marks)
 - b. Illustrate how different gear ratios can be obtained using planetary gear set in the automatic transmission system and also discuss the role of torque converter for automatic transmission system. (06marks)
 - c. Explain the construction and working principles of Hall effect sensor. (06marks)
- 3
 - a. The vehicle is to be started in a very cold winter morning and the it has to travel on hilly terrain. How does the engine cope up with different requirements as the vehicle goes from start to cruising? (08marks)
 - b. Justify the need of electronics in automotive vehicles, and also discuss the European and Indian emission standards. (06marks)
 - c. What are the design goals for hybrid vehicles? Compare the various system architectures for hybrid vehicles. (06marks)
- 4
 - a. Suggest and explain the suitable control system, which prevents the wheels from locking and enables safe braking. (08marks)
 - b. Calculate nominal and maximum T_{HEADER} , T_{RESPONSE} and T_{FRAME} , if LIN protocol is operating at 10 kbps baud rate and reserved time is set to 40% for transmitting four bytes of data. (06marks)
 - c. Explain the following with respect to FlexRay communication protocol
 - i) Bus level
 - ii) Bus gaurdian
 - iii) Communication cycle
 (06marks)
- 5
 - a. Suggest and explain the suitable control system, which deliver a better ride and handling experience over various driving conditions. (08marks)
 - b. If the vehicle longitudinal acceleration is zero. (06marks)
 - i) Calculate wheel slip, if vehicle speed is matching with wheel speed.
 - ii) Calculate wheel slip for front left and front right wheel, when the vehicle speed is 80 kmph and front left wheel speed and front right wheel speed sensor reading

are 71 kmph and 75 kmph respectively.

iii) Calculate wheel slip when panic braking is applied at 150 kmph and ABS is shut down due to pump failure.

- 6
- c. Draw and explain the MOST25 protocol frame format. (06marks)
 - a. The CAN node has to transmit the speed sensor information from engine ECU to dashboard ECU and the size of the information is 5 bits. Explain how this message is transmitted and discuss how the CAN receiver node determined whether the message is error free or not. (08marks)
 - b. Explain the suitable safety feature, which helps to prevent loss of control in curves and emergency steering maneuvers. (06marks)
 - c. Explain the frame format of CAN FD protocol. (06marks)

UNIT-III

- 7
- a. Assume that the driver has activated the cruise control switch for the desired speed of 80 kmph. Explain the suitable control algorithm to control the throttle angle for the following conditions
 - i) If the car traveling on a level road
 - ii) If the car is entering a long hill with a steady positive slope. (08marks)
 - b. What is ASIL? How to determine ASILs and ASIL levels. (06marks)
 - c. Explain the system architecture for autonomous vehicles. (06marks)
- 8
- a. Explain the need of on board and off-board diagnosis in automotive vehicles. (08marks)
 - b. Explain the analytical method for designing fault tolerant system with an example. (06marks)
 - c. Explain the architecture of UDS protocol. (06marks)